

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	264852	49.244923
1	[Git, Change, Create]	99583	18.515840
2	[Git, Change, Delete]	61316	11.400713
3	[Git, Change, Rename]	24820	4.614875
4	[File, Git]	16609	3.088173
5	[Git, Commit]	16390	3.047454
6	[Java, ByteCode, Parameter]	9457	1.758375
7	[Java, ByteCode, Member, Method]	9076	1.687535
8	[Java, ByteCode, Bound]	6545	1.216936
9	[Java, Value, ByteCode, Annotation]	5562	1.034163
10	[Java, ByteCode, ParameterizedType, Bound]	5029	0.935061
11	[Xml, Element]	2512	0.467066
12	[Java, ByteCode, Member, Field]	2233	0.415190
13	[Xml, Text]	1692	0.314600
14	[Java, ByteCode, Member, Constructor, Method]	1596	0.296750
15	[Java, ByteCode, Bound, WildcardType]	1079	0.200623
16	[Java, ByteCode, Bound, TypeVariable]	868	0.161390
17	[Java, ByteCode, Member, Method, Lambda]	839	0.155998
18	[Type, File, Java, ByteCode, JavaType]	698	0.129782
19	[Type, File, Java, Class, ByteCode]	675	0.125505
20	[Git, Change, Copy]	546	0.101520
21	[Json, Key]	523	0.097243
22	[Value, Json, Scalar]	513	0.095384
23	[Java, ByteCode, Member, GenericDeclaration, M...]	481	0.089434
24	[Type, File, Java, ByteCode, ResolvedDuplicate...]	477	0.088690
25	[Java, Value, ByteCode, Primitive]	364	0.067680
26	[Type, File, Java, ByteCode, ExternalType]	330	0.061358
27	[Author, Git, Person]	307	0.057082
28	[Committer, Git, Person]	248	0.046112
29	[Maven, Dependency]	205	0.038116

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

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Category	Percentage	Count
['Git', 'Update', 'Change']	49.24%	26,485,200
['Git', 'Change', 'Create']	18.52%	9,958,300
['Git', 'Change', 'Delete']	11.40%	6,131,600
['Git', 'Change', 'Rename']	4.61%	2,482,000
others	3.46%	1,858,700
['File', 'Git']	3.09%	1,660,900
['Git', 'Commit']	3.05%	1,639,000
['Java', 'ByteCode', 'Parameter']	1.76%	945,700
['Java', 'ByteCode', 'Member', 'Method']	1.69%	907,600
['Java', 'ByteCode', 'Bound']	1.22%	654,500
['Java', 'Value', 'ByteCode', 'Annotation']	1.03%	556,200
['Java', 'ByteCode', 'ParameterizedType', 'Bound']	0.94%	502,900

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000186
1	[Git, Branch]	1	0.000186
2	[Repository, File, Git]	1	0.000186
3	[File, Json]	2	0.000372
4	[File]	4	0.000744
5	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000744
6	[Maven, Exclusion]	6	0.001116
7	[Value, Array, Json]	8	0.001487
8	[Java, ByteCode, Member, GenericDeclaration, C...	9	0.001673
9	[Type, File, Java, ByteCode, Throwable, Extern...	10	0.001859
10	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002045
11	[Maven, Scm]	11	0.002045
12	[File, Maven, Xml, Pom, Document]	11	0.002045
13	[Type, File, Java, ByteCode, Void]	11	0.002045
14	[Java, ManifestSection]	11	0.002045
15	[File, Java, Manifest]	11	0.002045
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002045
17	[File, Java, ServiceLoader]	13	0.002417
18	[Maven, PluginExecution]	13	0.002417
19	[File, Java, Properties]	13	0.002417
20	[Maven, ExecutionGoal]	13	0.002417
21	[Type, File, Java, ByteCode, Enum]	17	0.003161
22	[jQAssistant, Rule, Concept]	19	0.003533
23	[Maven, Configuration]	20	0.003719
24	[Maven, Plugin]	20	0.003719
25	[Xml, Attribute]	22	0.004091
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.007251
27	[Type, File, Java, Class, ByteCode, Throwable]	42	0.007809
28	[Type, File, Java, ByteCode, Annotation]	42	0.007809
29	[Xml, Namespace]	44	0.008181

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

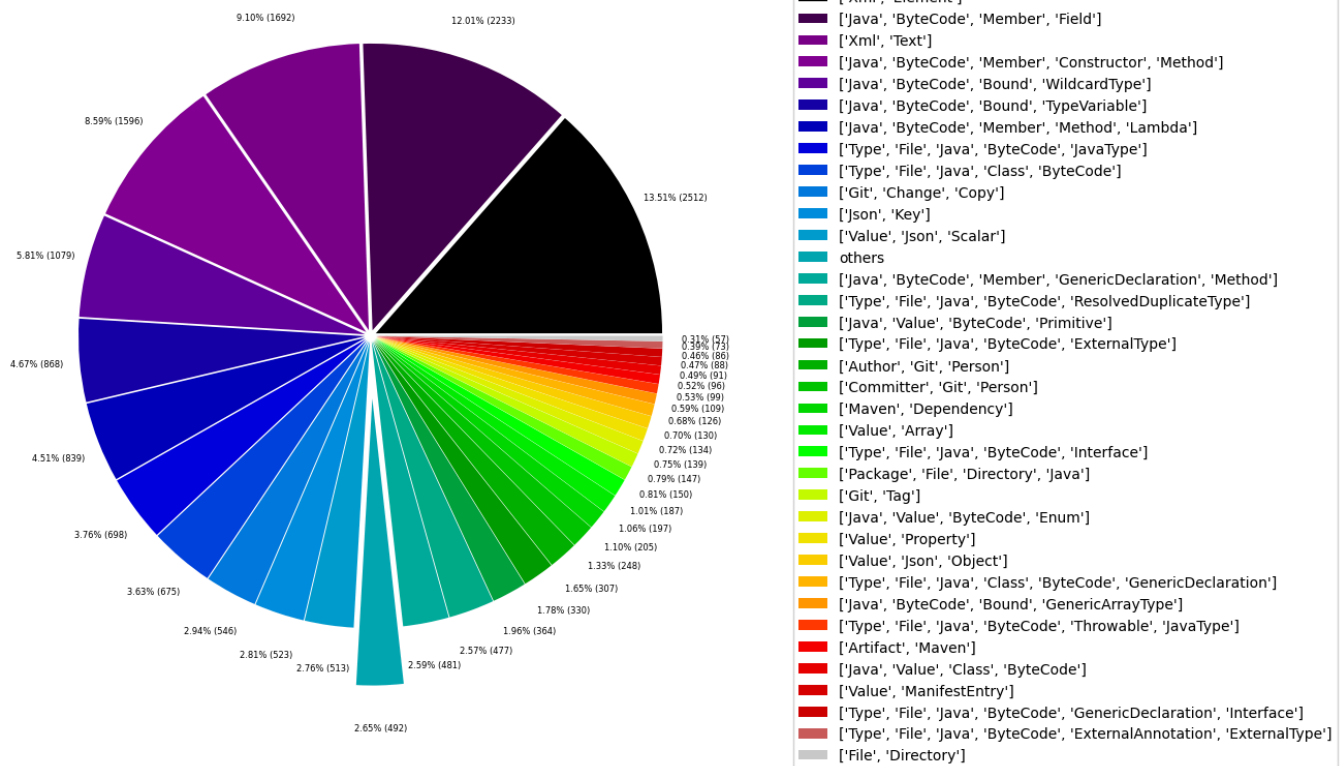


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

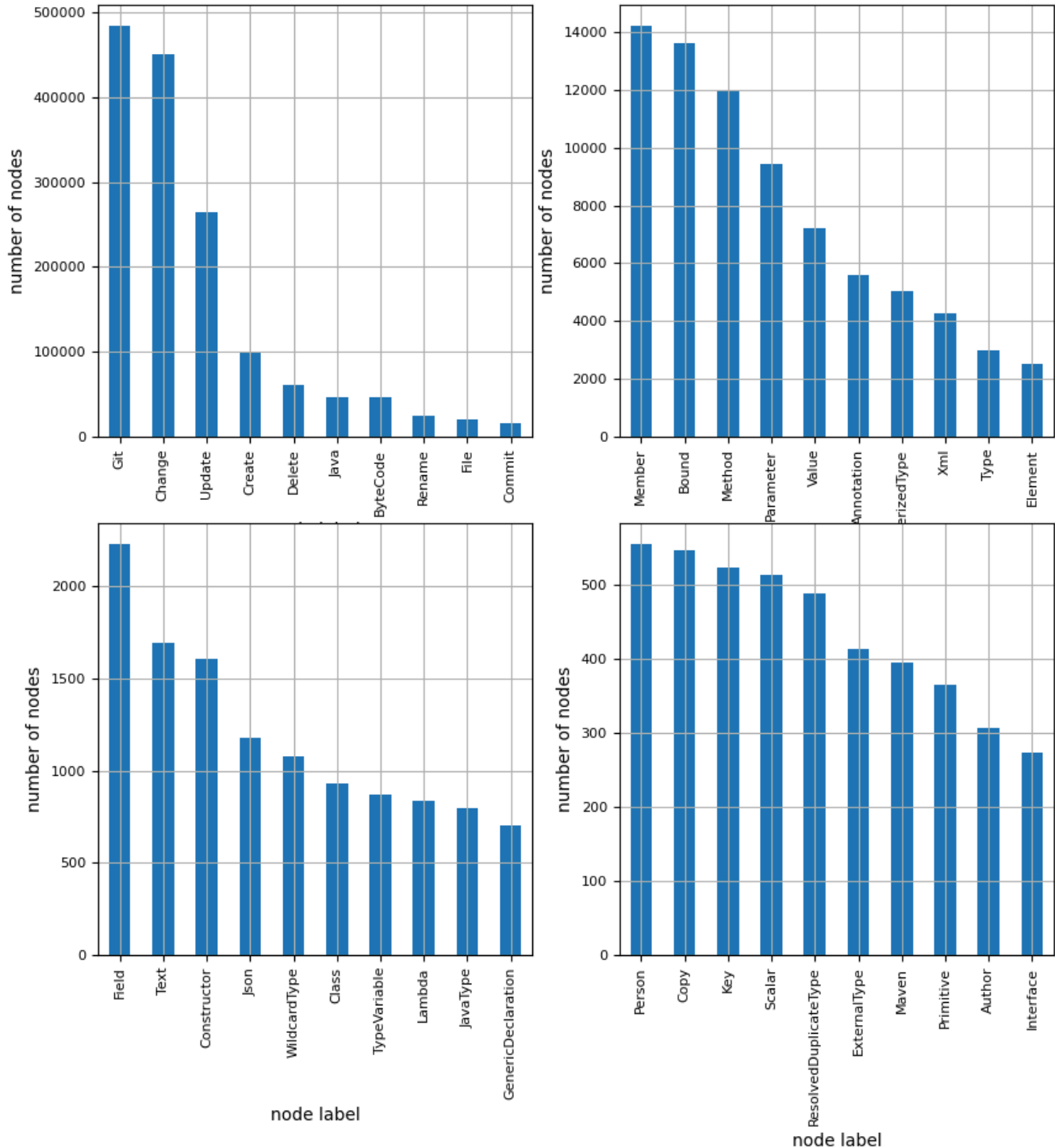
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	484820	90.144396
1	Change	451117	83.877871
2	Update	264852	49.244923
3	Create	99583	18.515840
4	Delete	61316	11.400713
5	Java	46665	8.676598
6	ByteCode	46456	8.637738
7	Rename	24820	4.614875
8	File	19861	3.692830
9	Commit	16390	3.047454
10	Member	14234	2.646581
11	Bound	13630	2.534277
12	Method	12001	2.231391
13	Parameter	9457	1.758375
14	Value	7226	1.343557
15	Annotation	5604	1.041973
16	ParameterizedType	5029	0.935061
17	Xml	4281	0.795982
18	Type	2979	0.553897
19	Element	2512	0.467066
20	Field	2233	0.415190
21	Text	1692	0.314600
22	Constructor	1605	0.298424
23	Json	1176	0.218658
24	WildcardType	1079	0.200623
25	Class	934	0.173662
26	TypeVariable	868	0.161390
27	Lambda	839	0.155998
28	JavaType	797	0.148189
29	GenericDeclaration	706	0.131269
30	Person	555	0.103193
31	Copy	546	0.101520
32	Key	523	0.097243
33	Scalar	513	0.095384
34	ResolvedDuplicateType	488	0.090736
35	ExternalType	413	0.076791
36	Maven	395	0.073444
37	Primitive	364	0.067680
38	Author	307	0.057082
39	Interface	273	0.050760

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1668493

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	451117	27.037392
1	MODIFIES	451117	27.037392
2	UPDATES	264852	15.873726
3	CREATES	124949	7.488734
4	DELETES	86136	5.162503
5	COMMITTED	32780	1.964647
6	RENAMES	24820	1.487570
7	INVOKES	22695	1.360209
8	HAS_PARENT	19906	1.193053
9	OF_TYPE	19011	1.139411
10	DEPENDS_ON	18178	1.089486
11	HAS_FILE	16609	0.995449
12	HAS_COMMIT	16390	0.982324
13	DECLARES	14588	0.874322
14	OF_RAW_TYPE	13531	0.810971
15	HAS	10124	0.606775
16	HAS_NEW_NAME	9599	0.575310
17	RETURNS	8301	0.497515
18	READS	6117	0.366618
19	HAS_ACTUAL_TYPE_ARGUMENT	5936	0.355770
20	ANNOTATED_BY	5561	0.333295
21	CHANGED_TOGETHER_WITH	4683	0.280672
22	OF_GENERIC_TYPE	4539	0.272042
23	SIMILAR	3496	0.209530
24	CONTAINS	3309	0.198323
25	RETURNS_GENERIC	2679	0.160564
26	RESOLVES_TO	2676	0.160384
27	HAS_FIRST_CHILD	2512	0.150555
28	HAS_LAST_CHILD	2512	0.150555
29	HAS_ELEMENT	2490	0.149236

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

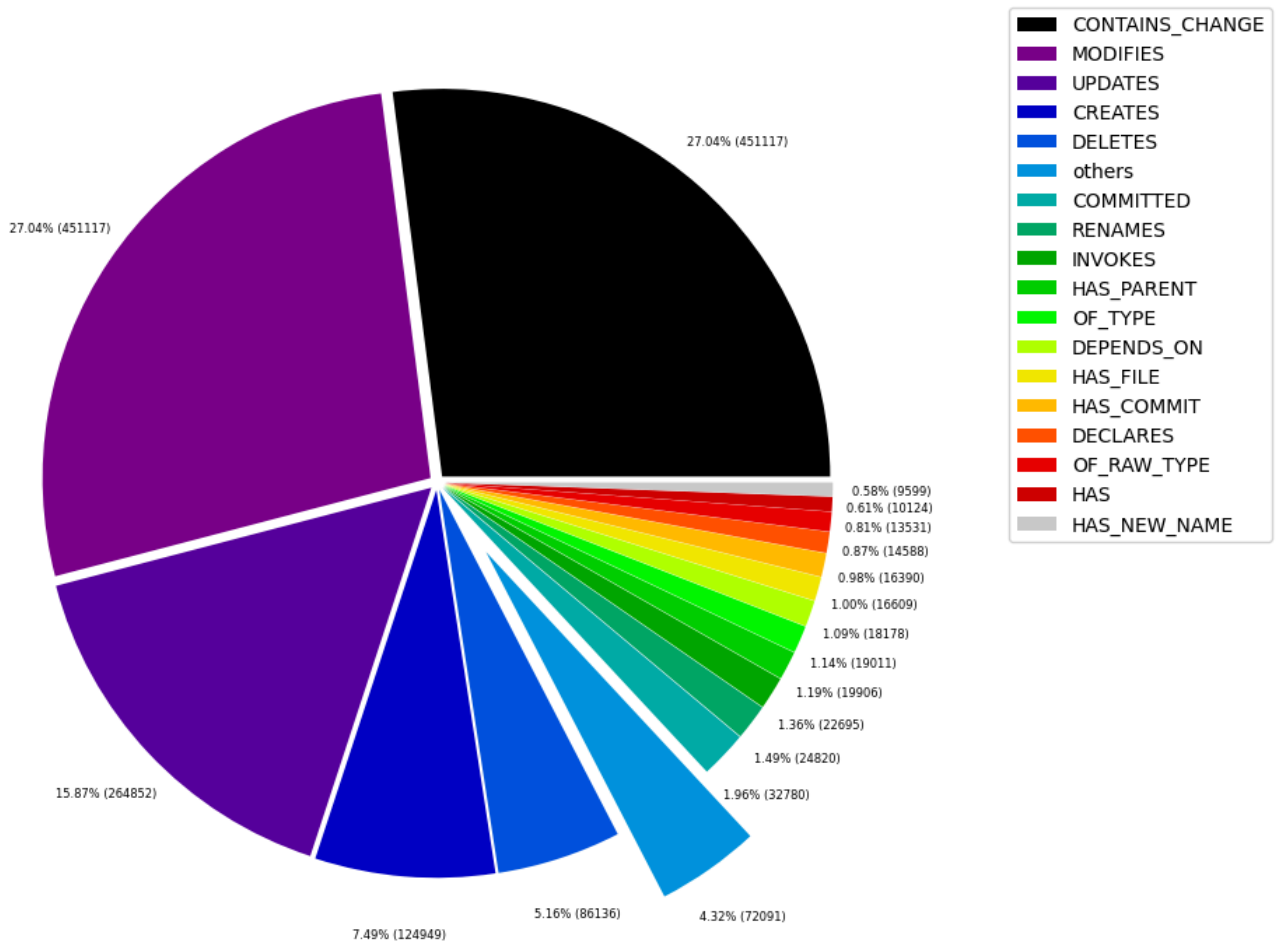


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

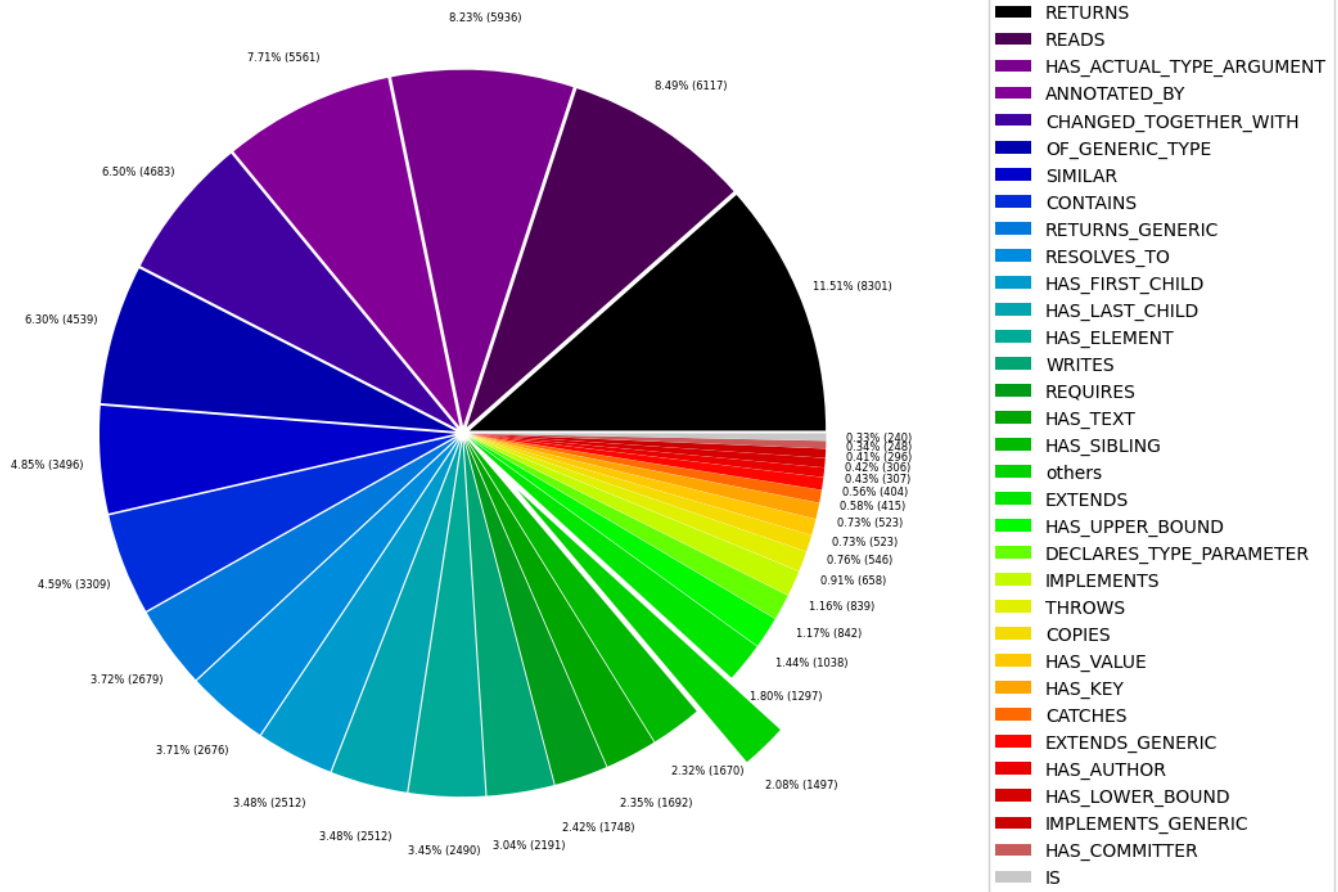
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000060
1	HAS_BRANCH	1	0.000060
2	HAS_HEAD	2	0.000120
3	THROWS_GENERIC	5	0.000300
4	EXCLUDES	6	0.000360
5	HAS_SCM	11	0.000659
6	DESCRIBES	11	0.000659
7	HAS_GOAL	13	0.000779
8	HAS_EXECUTION	13	0.000779
9	HAS_ROOT_ELEMENT	14	0.000839
10	INCLUDES_CONCEPT	19	0.001139
11	IS_ARTIFACT	20	0.001199
12	HAS_CONFIGURATION	20	0.001199
13	USES_PLUGIN	20	0.001199
14	OF_NAMESPACE	22	0.001319
15	HAS_ATTRIBUTE	22	0.001319
16	REQUIRES_TYPE_PARAMETER	26	0.001558
17	REQUIRES_CONCEPT	28	0.001678
18	DECLARES_NAMESPACE	44	0.002637
19	HAS_DEFAULT	54	0.003236
20	HAS_COMPONENT_TYPE	109	0.006533
21	CONTAINS_VALUE	126	0.007552
22	HAS_TAG	147	0.008810
23	ON_COMMIT	147	0.008810
24	TO_ARTIFACT	205	0.012287
25	DECLARES_DEPENDENCY	205	0.012287
26	COPY_OF	206	0.012346
27	IS	240	0.014384
28	HAS_COMMITTER	248	0.014864
29	IMPLEMENTS_GENERIC	296	0.017741

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Update, Change]	MODIFIES	[File, Git]	264852		264852
1	[Git, Update, Change]	UPDATES	[File, Git]	264852		264852
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	264852		16390
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	99583		16390
4	[Git, Change, Create]	MODIFIES	[File, Git]	99583		99583
5	[Git, Change, Create]	CREATES	[File, Git]	99583		99583
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	61316		16390
7	[Git, Change, Delete]	MODIFIES	[File, Git]	61316		61316
8	[Git, Change, Delete]	DELETES	[File, Git]	61316		61316
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	24820		16390
10	[Git, Change, Rename]	RENAMES	[File, Git]	24820		24820
11	[Git, Change, Rename]	DELETES	[File, Git]	24820		24820
12	[Git, Change, Rename]	CREATES	[File, Git]	24820		24820
13	[Git, Change, Rename]	MODIFIES	[File, Git]	24820		24820
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	19895		16390
15	[Repository, File, Git]	HAS_FILE	[File, Git]	16609		1
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	16390		1
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	16390		307
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	16390		248
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13552		9046
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9599		16609
21	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5531		9046
22	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5443		9046
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	5201		5562
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	4218		9457
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3915		9457
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3885		16609
27	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2884		5029
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2490		2512
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2376		6545

Graph Density

```
total_number_of_nodes (vertices): 537826
total_number_of_relationships (edges): 1668493
-> total directed graph density: 5.768217259182443e-06
-> total directed graph density in percent: 0.0005768217259182444
```